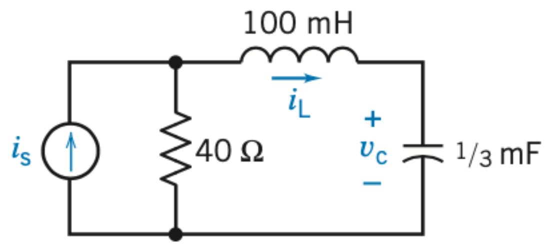
Problem #1

Find the characteristic equation and its roots for the circuit

Answer: $s^2 + 400s + 3 \times 10^4 = 0$

roots: $s = -300, \; -100$

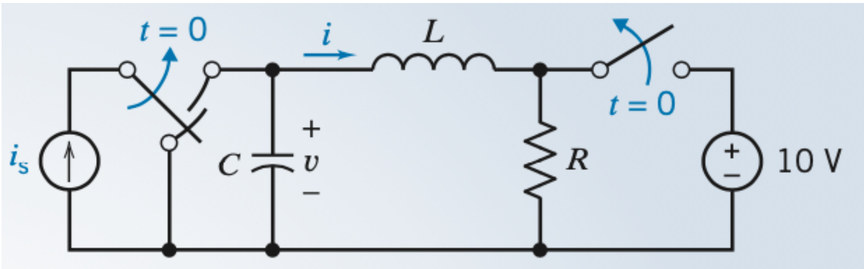


For the following problems: Steps or Method of Circuit Analysis

1. Identify the state variables as the independent capacitor voltages and inductor currents.
2. Determine the initial conditions at $t = 0$ for the capacitor voltages and the inductor currents.
3. Obtain a first-order differential equation for each state variable, using KCL or KVL.
4. Use the operator s to substitute for d=dt.
5. Obtain the characteristic equation of the circuit by noting that it can be obtained by setting the determinant of Cramer's rule equal to zero.
6. Determine the roots of the characteristic equation, which then determine the form of the natural response.
7. Obtain the second-order (or higher-order) differential equation for the selected variable x by Cramer's rule.
8. Determine the forced response x_f by assuming an appropriate form of x_f and determining the constant by substituting the assumed solution in the second-order differential equation.
9. Obtain the complete solution $x = x_n + x_f$.
10. Use the initial conditions on the state variables along with the set of first-order differential equations (step 3) to obtain
    dx(0)=dt.
11. Using x(0) and dx(0)=dt for each state variable, find the arbitrary constants $A_1, A_2, \ldots A_n$ to obtain the complete solution x(t).

Problem #2

Find i(t) for t > 0 for the circuit shown in figure below when R = 3 ohm, L = 1H, C=1/2F, and $i_S = 2e^{3t}$ A. Assume steady state at $t = 0^-$.



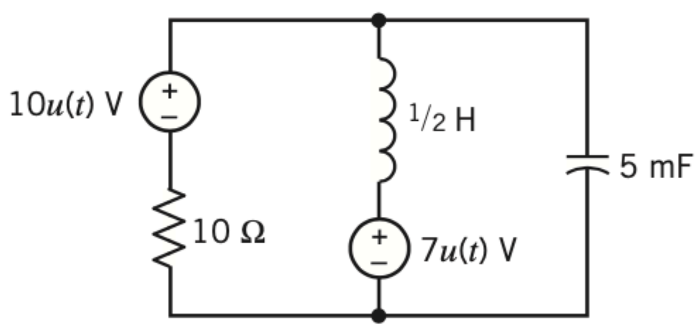
$$i = 12e^{-t} - 14e^{-2t} + 2e^{-3t}\ \mathrm{A}$$

Problem #3

German automaker Volkswagen, in its bid to make more efficient cars, were first to come up with an auto whose engine saves energy by shutting itself off at stoplights. The stop–start system springs from a campaign to develop cars in all its world markets that use less fuel and pollute less than vehicles now on the road. The stop–start transmission control has a mechanism that senses when the car does not need fuel: coasting downhill and idling at an intersection. The engine shuts off, but a small starter flywheel keeps turning so that power can be quickly restored when the driver touches the accelerator.

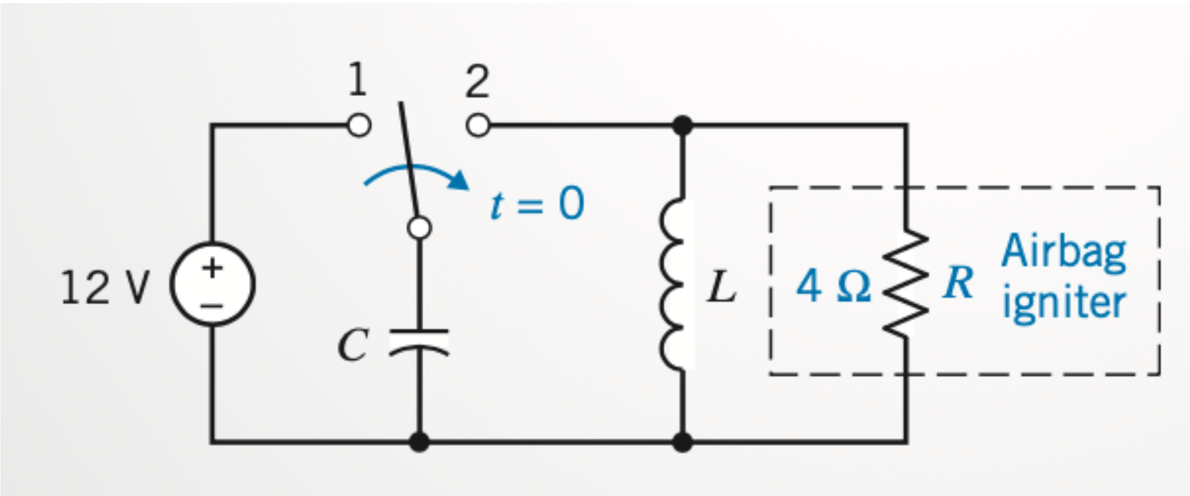
A model of the stop–start circuit is shown in figure below. Your job is to make a circuit better, with cheaper parts… But before that you need to find out what the competition Volkswagen is doing. Determine the characteristic equation and the natural frequencies for the circuit.

Answer: $s^2 + 20s + 400 = 0$
$s = -10 \pm j17.3$



Problem #4

Airbags are widely used for driver and passenger protection in automobiles. A pendulum is used to switch a charged capacitor to the inflation ignition device, as shown in Figure below. The automobile airbag is inflated by an explosive device that is ignited by the energy absorbed by the resistive device represented by R. To inflate, it is required that the energy dissipated in R be at least 1 J. It is required that the ignition device trigger within 0.1 s. Select the L and C that meet the specifications.



(Hint):

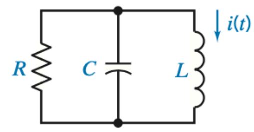
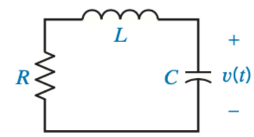
Describe the Situation and the Assumptions

1. The switch is changed from position 1 to position 2 at t = 0.
2. The switch was connected to position 1 for a long time.
3. A parallel RLC circuit occurs for t >= 0.

Select L and C so that the energy stored in the capacitor is quickly (read less than 0.1s) delivered to the resistive device R.

Ans. C = 1/16 F L = 0.065 H

Review Summary

	PARALLEL *RLC*	SERIES *RLC*
Circuit		
Differential equation	$\frac{d^2}{dt^2}i(t)+\frac{1}{RC}\frac{d}{dt}i(t)+\frac{1}{LC}i(t)=0$	$\frac{d^2}{dt^2}v(t)+\frac{R}{L}\frac{d}{dt}v(t)+\frac{1}{LC}v(t)=0$
Characteristic equation	$s^2+\frac{1}{RC}s+\frac{1}{LC}=0$	$s^2+\frac{R}{L}s+\frac{1}{LC}=0$
Damping coefficient, rad/s	$\alpha=\frac{1}{2RC}$	$\alpha=\frac{R}{2L}$
Resonant frequency, rad/s	$\omega_0=\frac{1}{\sqrt{LC}}$	$\omega_0=\frac{1}{\sqrt{LC}}$
Damped resonant frequency, rad/s	$\omega_d=\sqrt{\left(\frac{1}{2RC}\right)^2-\frac{1}{LC}}$	$\omega_d=\sqrt{\left(\frac{R}{2L}\right)^2-\frac{1}{LC}}$
Natural frequencies: overdamped case	$s_1,s_2=-\frac{1}{2RC}\pm\sqrt{\left(\frac{1}{2RC}\right)^2-\frac{1}{LC}}$ when $R<\frac{1}{2}\sqrt{\frac{L}{C}}$	$s_1,s_2=-\frac{R}{2L}\pm\sqrt{\left(\frac{R}{2L}\right)^2-\frac{1}{LC}}$ when $R>2\sqrt{\frac{L}{C}}$
Natural frequencies: critically damped case	$s_1=s_2=-\frac{1}{2RC}$ when $R=\frac{1}{2}\sqrt{\frac{L}{C}}$	$s_1=s_2=-\frac{R}{2L}$ when $R=2\sqrt{\frac{L}{C}}$
Natural frequencies: underdamped case	$s_1,s_2=-\frac{1}{2RC}\pm j\sqrt{\frac{1}{LC}-\left(\frac{1}{2RC}\right)^2}$ when $R>\frac{1}{2}\sqrt{\frac{L}{C}}$	$s_1,s_2=-\frac{R}{2L}\pm j\sqrt{\frac{1}{LC}-\left(\frac{R}{2L}\right)^2}$ when $R<2\sqrt{\frac{L}{C}}$